TC updates and ideas

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1.**TC updates since 0x14**
2.**Some issues**
3.**Some ideas**

Major TC updates

- HTB HW offloading (from Maxim Mikityanskiy)
- Qevent block support (from Petr Machata)
- IP fragmentation support (from wenxu)
- Lockless Qdisc fixes (from Yunsheng Lin)

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HTB HW offloading

- Get rid of qdisc tree lock contention
- Similar to sch_mq, a leaf class maps to HW queue
- Relies on clsact to do filtering
- Ratelimit in HW

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Qevent block

- Attach TC filter and action in other places than enqueue
- For example, RED congestions, early drops
- tc qdisc add dev eth0 root handle 1: red limit 500K avpkt 1K qevent early block 10
- tc filter add block 10 matchall action mirred egress mirror dev eth1

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IP fragmentation support

- Handle over-MTU size packets for mirred action
- Call into IP fragmentation layer
- Hijack dev_queue_xmit()

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Lockless qdisc

- Lockless is hard, race condition is hard to verify
- Qdisc scheduling is hard without lock
- Should we move qdisc lock down to each Qdisc? And Qdisc scheduling?

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Some issues

- pfifo_fast is getting uglier
- qdisc tree lock contention
- TC action init API is very ugly
- Datapath tests for tc-testing

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Some ideas

- eBPF qdisc (still WIP)
- A complete HTB replacement (vs. EDT)
- Backpressure
- Standalone TC filters?
- Ingress queues?
- Ingress traffic shaping
- eBPF iterator support?

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THANKS

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